Homepage Google Scholar
+1 520-501-4268 xinyuguo1226@gmail.com
github/xinnyuguo

# Xinyu Guo Ph.D. student

## Education

**University of Arizona** Arizona, United States
*Ph.D. Student in Computer Science* 2024 - Present
- Advisor: Prof. Mihai Surdeanu
- Research Interests: LLM Reasoning, RL Credit Assignment, Reinforcement Learning, Interpretability

**Shandong University** Shandong, China
*B.E. in Computer Science* 2019 - 2023
- GPA: 89.56/100.00
- Graduated with Honors

## Research Overview

My research focuses on **LLM-based reasoning**, along two directions:

- **RL Credit Assignment for Reasoning Supervision**: outcome rewards tell a model *whether* it reasoned well, but not *where*. I assign credit at a finer granularity: to the key phrases in a reasoning trace, using rewards derived from the model's own correct predictions (CogRE), and to the tokens responsible for each rubric criterion, using rubric-conditioned critics (DVE).
- **Reasoning Mechanisms**: looking inside LLMs to understand how concepts and their logical structure support reasoning, by aligning sparse autoencoder features with concepts (AlignSAE) and taxonomies (TaxoSAE).

## Publications

***RL Credit Assignment for Reasoning Supervision***

[2026] **Xinyu Guo**, Zhengliang Shi, Minglai Yang, Mahdi Rahimi, Mihai Surdeanu*. The Answer Lies Within: Self-Derived Rewards Enable Explainable Relation Extraction. ***EMNLP 2026 Main.***
*University of Arizona, Shandong University*.

[2026] **Xinyu Guo**, Zhengliang Shi, Minglai Yang, Chen Xu, Lingyong Yan. Disentangled Value Estimation: Rethinking Scalar Critics in Rubric-Based RL. *Under Review*.

***Reasoning Mechanisms***

[2026] **Xinyu Guo**, Minglai Yang, Zhengliang Shi, Clayton T. Morrison, Eduardo Blanco, Mihai Surdeanu. TaxoSAE: Rethinking Sparse Autoencoders through Taxonomic Logic. *Under Review*.

[2026] Minglai Yang, **Xinyu Guo**, Zhengliang Shi, Jinhe Bi, Steven Bethard, Mihai Surdeanu, Liangming Pan. AlignSAE: Concept-Aligned Sparse Autoencoders. ***TMLR 2026***.
*University of Arizona, Shandong University, Peking University, LMU Munich*.

***Other Publications***

[2026] Minglai Yang, **Xinyu Guo**, Utkarsh Tyagi, Mian Zhang, Razvan Dumitru, Sunjie Hou, Yunzhong He, Daniel Yue Zhang, Ying Liu. Rubric Dropout: A Simple Way to Mitigate Reward Hacking in Rubric-as-Reward RL. *arXiv 2026*.
*Scale AI, University of Arizona, University of Texas at Dallas*.

[2026] Minglai Yang, **Xinyu Guo**, Renxiong Wang, Daniel Yue Zhang, Yunzhong He. From Copying to Composition: How Language Models Develop In-Context Learning. *Under Review*.

[2026] Minglai Yang, Xinyan Velocity Yu, Pengyuan Li, **Xinyu Guo**, Zhenting Qi, ..., Zexue He. Dr. DocBench: A Comprehensive Benchmark for Expert-Level and Difficult Document Parsing. *NeurIPS 2026 Workshop LCFM*.

[2026] Yunze Xiao, Jienan Chen, **Xinyu Guo**, Minghao Liu, Zihan Wang, Weihao Xuan. RecuritOpsBench: When Agents Fail to Ask for the Information They Need. *Under Review*.

[2025] Zhengliang Shi, ..., **Xinyu Guo**, ..., Zhaochun Ren. Deep Research: A Systematic Survey. *arXiv 2025*.
*Shandong University, University of Arizona, CMU, Tsinghua University, etc.*

[2025] Zhaoze Sun, Chengliang Chai*, Qiyan Deng, Kaisen Jin, **Xinyu Guo**, Han Han, Ye Yuan, Guoren Wang, Lei Cao. QUEST: Query Optimization in Unstructured Document Analysis. Accepted by *VLDB* 2025.
*University of Arizona, MIT, Beijing Institute of Technology*.

## Internships

**Abaka AI** | Mountain View, CA, United States 2026.05 - 2026.10
- Role: Research Scientist Intern
- Built RL environments and post-training data for tool-using customer-service agents on $\tau^2$-bench, and trained call agents with RL (RetailServe).
- Contributed to Dr. DocBench and helped organize the DocInsights 2026 workshop at EMNLP 2026.

## Research Experience

**CLU-LAB, University of Arizona** | Arizona, United States 2025.03 - Present
- Advisor: Prof. Mihai Surdeanu
- Research Topic: Natural Language Processing, Reinforcement Learning, LLMs.

**CAES Research Center, Shandong University** | Shandong, China 2023.09 - 2024.09
- Advisor: Prof. Zhongxing Yu
- Research Topic: Program Analysis, Automated Program Repair, AI4SE.

**Oxford Winter Exchange Program, Oxford University** | UK 2022.02
- Advisor: Prof. Jeremy Pitt
- Topic: AI Myths, Reality, and Human-AI Interaction.

## Grant and Services

**Co-PI for ABC SEED Grant, Arizona Astrobiology Center**

**Reviewer for ARR 2026**

## Awards and Honors

**Galileo Circle Scholarship**, University of Arizona 2026

**Science Graduate College Fellowship**, University of Arizona 2024, 2025

**Outstanding Undergraduate Thesis Award**, Shandong University 2023

**Outstanding Graduate Student**, Shandong University 2023

**Academic Scholarship**, Shandong University 2022

**The 18th Certified Software Professional Examination Top 4.57%**, CCF 2021

## Skills

**Programming Languages**: `Python`, `C/C++`, `MATLAB`, `Go`, `Java`, `HTML`.

**Libraries & Framework**: `PyTorch`, `Numpy`, `vLLM`, `VeRL(RL)`, `Hugging Face Transformers`, `SQLite`, `MySQL`, `Spring Boot`, `JavaScript`, `Qt`.

## Teaching

**Teaching Assistant for CSC337 Web Programming, CSC460 Database**
*University of Arizona, US* 2024, 2025

**Teaching Assistant for Software Engineering**
*Shandong University, China* 2023